Assignment 1: Report

November 5, 2026

1

1.1 MLP

Started with a baseline model with hidden_size = 256, dropout = 0.2, optimizer = SGD, and learning_rate = 0.01. Used a batch of 64 and 10 epochs. Here, we varied the activation layers. Tried out ReLU, LeakyReLU, and GELU. Using a 90/10 split for training and validation, LeakyReLU resulted in the highest validation accuracy $0.\bar{9}018$.

Will continue varying the optimizers, using LeakyReLU as the activation function. Tried out SGD, Adam, and AdamW. Again using a 90/10 split for training and validation, AdamW resulted in the highest validation accuracy 0.8650.

Tried out learning rate [0.001, 0.01, 0.1]. Learning rate $0.\bar{0}1$ resulted in the highest validation accuracy of 0.9025.

Tried out hidden sizes of 128, 256, and 512. Hidden size of 128 was the best, with a validation accuracy of 0.8722.

Tried out dropout values of 0.0, 0.2, and 0.5. Dropout $0.\bar{0}$ resulted in the highest validation accuracy of 0.8788.

1.2 CNN

We started by setting seeds for reproducibility, and then importing the MNIST dataset from `torchvision.datasets`. After importing the train and test separately, we split the training dataset into two subsets: what we will use for training (90%), and validation (10%). Setting batch size to 64, we created data loaders that will then load the data into the models.

We set up a basic convolutional neural network, similar to the one depicted in Figure 1, with two convolutional layers and max pooling between, with a customisable activation function.

Defining functions for training, validating, and testing, we then moved to the systematic testing of parameters.

We tested different activation functions:

ReLU

$$f(x) = \begin{cases} x & x \geqslant 0 \\ 0 & \text{otherwise} \end{cases}.$$

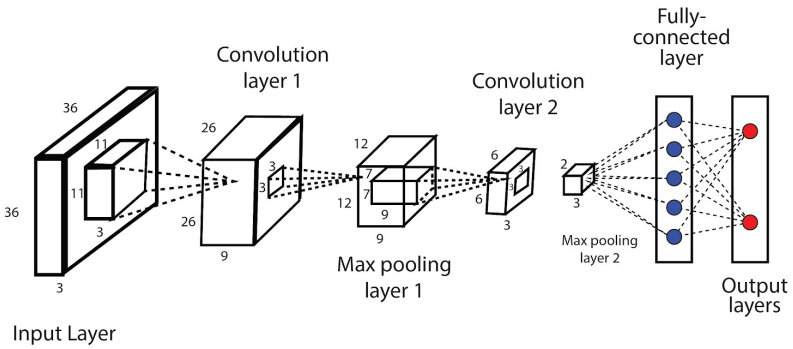


Figure 1: Sample Convolutional Neural Network

Leaky ReLU

$$f(x) = \begin{cases} x & x \geqslant 0 \\ mx & \text{otherwise} \end{cases}.$$

GeLU

$$f(x) = x \cdot \Phi(x).$$